

Evidence for variability of atmospheric hydroxyl radicals over the past quarter century

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[1] The hydroxyl free radical (OH) is the major oxidizing chemical in the atmosphere, destroying about 3.7 petagrams (Pg) of trace gases each year, including many gases involved in ozone depletion, the greenhouse effect and urban air pollution. Measurements of 1,1,1-trichloroethane (methyl chloroform, CH_3CCl_3), which reacts with OH, provide the most accurate method currently utilized for determining the global behavior of OH. We report that CH_3CCl_3 levels rose steadily from 1978 to reach a maximum in 1992 and have since decreased rapidly to levels in 2004 about 30% of the levels when measurements began in 1978. Analysis of these observations shows that global average OH levels had a small maximum around 1989 and a larger minimum around 1998, with OH concentrations in 2003 being comparable to those in 1979. This post-1998 recovery of OH reported here contrasts with the situation 4 years ago when reported OH was decreasing. The 1997–1999 OH minimum coincides with, and is likely caused by, major global wildfires and an intense El Niño event at this time. **Citation:** Prinn, R. G., et al. (2005), Evidence for variability of atmospheric hydroxyl radicals over the past quarter century, *Geophys. Res. Lett.*, 32, L07809, doi:10.1029/2004GL022228.

1. Introduction

[2] The hydroxyl free radical is measurable directly at the local scale, but presently cannot be measured directly at the regional to global scale [Ehhalt, 1999]. Large-scale OH concentrations and trends can however be inferred indirectly from long-term global measurements and emission estimates of CH_3CCl_3 , since OH is the major destruction mechanism for this chemical [Prinn et al., 1992, 2001; Krol and Lelieveld, 2003; Montzka et al., 2000]. The potentials for global warming and stratospheric ozone depletion of a large number of chemicals can then be estimated from the derived OH concentrations [Montzka et al., 2003].

[3] CH_3CCl_3 has been measured continuously from July 1978 to present in three sequential experiments: the Atmospheric Lifetime Experiment (ALE), the Global Atmospheric Gases Experiment (GAGE) and the Advanced Global Atmospheric Gases Experiment (AGAGE) [Prinn et al., 2000, 2001]. Figure 1 shows monthly-mean dry-air mole fractions (χ) and standard deviations (σ) computed from the approximately 120 (ALE), 360 (GAGE), and 1080 (AGAGE) measurements made each month at each station in a five-station network. The observed variations in χ are caused by the early increase and later decrease in CH_3CCl_3 emissions which result from the United Nations Montreal Protocol [Prinn et al., 2001; Montzka et al., 2003]. They are also influenced by the varying distances from the mainly Northern Hemisphere midlatitude sources, the rate and seasonal variations in global circulation, and the seasonal oscillations in the rate of the reaction of OH with CH_3CCl_3 which has a local summer maximum [Prinn et al., 1992].

2. Emissions

[4] Emission estimates for CH_3CCl_3 have been determined traditionally from global and regional sales and end-use data from industry and, more recently, from consumption data collected under the United Nations Montreal Protocol [McCulloch and Midgley, 2001]. We combine these estimates with independent emission estimates based on recent measurements of polluted air in industrial regions to define a “reference” emission scenario. The 2000–2003 emissions for Europe have been estimated using concurrent observations from Switzerland and Ireland (AGAGE) [Reimann et al., 2005], suggesting that about 2.0, 2.3, 1.4 and 1.2 Gg year^{−1} respectively need to be added over this 4-year period to the above industry/United Nations estimates for Europe. Estimates of European emissions in 2000 exceeding 20 Gg [Krol et al., 2003] based on aircraft data are not evident from the above extensive surface station data [Reimann et al., 2005], and are not therefore considered in our reference emissions. Emissions per person estimated for the east coast U.S.A. using Massachusetts observations [Barnes et al., 2003] are about 1.8 times greater than those estimated for the west coast U.S.A. using AGAGE California observations (J. Li et al., Halocarbon emissions estimated from AGAGE measured pollution events at Trinidad Head, CA, submitted to *Journal of Geophysical Research*, 2005) (see auxiliary material¹). Combining these east and west coast estimates, multiplying by the U.S.A.

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population, and then subtracting industry/United Nations estimates, implies that 1996–1998 U.S. emissions may be underestimated by industry/United Nations data by on average about 9.0 Gg year^{-1} over this 3-year period. *Millet and Goldstein* [2004] use east and west coast observations to infer 2002 U.S.A. emissions of about 3 Gg year^{-1} (their larger emission estimates in earlier years are based only on the *Barnes et al.* [2003] east coast study). East Asian emissions deduced from aircraft data in 2001 are about 1.7 Gg above industry/United Nations data [Palmer et al., 2003]. In contrast, recent Australian emissions are negligible [Prinn et al., 2001], and Russian emissions in 2001 were undetectable [Hurst et al., 2004].

[5] We could explain approximately these additional regional emissions in the stated years (and presumed similar additional emissions in nearby years and other countries) by placing 5% of annual CH_3CCl_3 sales into a category where it is emitted at a constant rate over the subsequent 10 years. This provides additional emissions of 18.8, 18.3, 15.4, 12.3, 8.9, 6.4, and 3.6 Gg/year in 1996–2002. Alternatively, we could attribute these additional European, East Asian and U.S. emissions to illegal imports from legal producers and consumers elsewhere in the Northern Hemisphere who report their production and consumption under the Montreal Protocol. In this case the industry/United Nations estimates would not change for our global OH estimates. Therefore, we use the average of the industry/United Nations emissions and “5% delayed” industry emissions as our reference and increase the (1σ) industry-based error bars [McCulloch and Midgley, 2001] to include both the industry and 5%-delayed industry estimates where necessary (see auxiliary material).

3. Inverse Method

[6] A recursive weighted least squares (Kalman) filter and a 2-dimensional (2D) global model are used to deduce CH_3CCl_3 lifetimes (τ) and OH concentrations and trends [Prinn et al., 2001] (see auxiliary material). The 2D model is very flexible and computationally efficient compared to three-dimensional (3D) models. This enables multiple runs to examine the effects of model transport and chemistry errors on our OH determinations [Prinn et al., 2001]. Tests with a high-resolution 3D model with interannually varying and observationally constrained meteorology confirm that the χ and σ values at a particular station define well the large volume averages corresponding to the above 2D model (see auxiliary material). The 2D model includes a small oceanic CH_3CCl_3 sink [Yvon-Lewis and Butler, 2002]. It simulates well the magnitude of the stratospheric sink for CH_3CCl_3 [Montzka et al., 2003], and shows the expected changes in this sink over time [Krol and Lelieveld, 2003] (see auxiliary material).

[7] The basic approach is to multiply the “reference” OH concentrations in the 8 lower atmospheric boxes in the 2D model by a dimensionless factor (f) which is either held constant during each of the twenty-five 1-year intervals, twelve 2-year (plus one 1-year), or eight 3-year (plus one 2-year) intervals between 1979 and 2003, or expressed as a polynomial $f_p = a + bN P_1(t) + c(N^2/3)P_2(t) + d(N^3/15)P_3(t) + e(N^4/105)P_4(t)$. Here P_n is a Legendre polynomial of order n , and t is dimensionless time normalized to N and measured from the mid-point of the $2N$ -year-long 1979–2003 interval. The 9–25 f values, or the 5 unknown

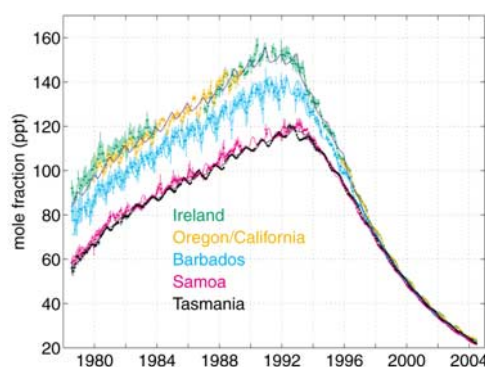


Figure 1. ALE/GAGE/AGAGE monthly mean dry-air mole fractions (dots) and standard deviations (error bars) for CH_3CCl_3 expressed as parts in 10^{12} (parts per trillion, ppt) from the five indicated stations. The measurements are on the Scripps Institution of Oceanography SIO-1998 absolute calibration scale [Prinn et al., 2000] (see auxiliary material). Significant intramonthly variations include elevated CH_3CCl_3 levels in polluted air from nearby industrial regions [Prinn et al., 1992, 2000]. To help ensure that χ and σ represent semi-hemispheric scales, periods of obvious pollution are omitted from their calculation [Prinn et al., 2000]. Also shown (solid curves) are the mole fractions computed using the optimally estimated 1-year average OH concentrations in the 2D model. The time coordinate refers to the beginning of each year in all figures.

f_p coefficients are then optimally estimated using each month's observations. We chose 2-year and 3-year as well as 1-year f averages to decrease possible errors due to the use of annually repeating circulation in our 2D model [Krol and Lelieveld, 2003]. The estimated f values are used finally to correct the prescribed “reference” OH values in the model.

4. Results

[8] Figure 2 shows the derived temporal variations in OH concentrations. Taking into account all measurement, calibration, emission and modeling errors, the anharmonic variation in the global trend is only marginally significant (the d and e values in f_p are borderline statistically nonzero; see auxiliary material). Figure 2 also shows the results of using the McCulloch and Midgley [2001] emissions instead of the reference emissions. As expected, the OH estimates are higher in the early years and lower in the later years, but the interannual variations remain. Using these OH estimates, the derived time-averaged CH_3CCl_3 lifetimes (atmospheric content divided by loss rate) in years are 4.9 ± 0.3 (total loss), $6.0^{+0.5}_{-0.4}$ (loss via tropospheric OH), 38^{+15}_{-11} (stratospheric loss), and 94^{+51}_{-11} (oceanic loss). The derived CH_4 lifetimes are $9.3^{+0.7}_{-0.6}$ (total loss), $10.2^{+0.9}_{-0.7}$ (tropospheric OH loss), and 110 (stratospheric loss) years respectively (see auxiliary material).

[9] Wennberg et al. [2004] have proposed that the polar oceans may have stored methyl chloroform during the pre-1992 years when its atmospheric levels were rising, but began re-emitting it in the subsequent years, thus lessening the overall oceanic sink [see also Krol and

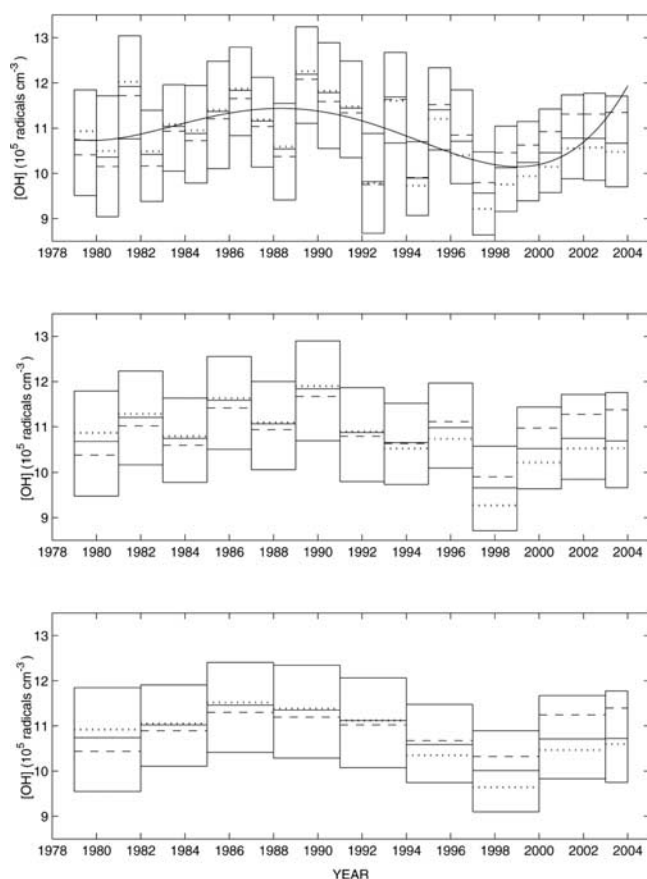


Figure 2. One-, two- and three-year weighted-average estimated OH concentrations with 1σ error boxes (excluding rate constant errors). These absolute concentrations, but not the relative OH variations, depend on the weighting and model used (see auxiliary material). Errors (ϵ) due to random measurement (instrumental, and model grid - measurement site mismatch) errors (σ) are automatically calculated in the Kalman Filter. Errors due to uncertainties in model transport, model chemical parameters, emissions and absolute calibration, have been calculated using 10,000-run Monte Carlo approaches and are subsequently added to ϵ [Prinn *et al.*, 2001; McCulloch and Midgley, 2001] (see auxiliary material). Also shown is the OH defined using f_p with optimally-estimated coefficients (solid line; this line is not intended for forecasting), and the results when the reference emissions are replaced by the McCulloch and Midgley [2001] industry emissions (dotted lines), and when the oceanic sink based on Yvon-Lewis and Butler [2002] is replaced by the computed fluxes of Wennberg *et al.* [2004] (dashed lines).

Lelieveld, 2003]. When we replaced our oceanic removal parameterization [Yvon-Lewis and Butler, 2002] with their computed latitude-varying oceanic fluxes, we estimated higher OH concentrations after 1994 but the interannual changes remain (see Figure 2).

[10] To test the influence of any one station on our conclusions about multi-year OH variations, we repeated our estimations with and without the use of each station's data, especially Samoa which is sensitive to ENSO before 1997 when the north-south CH_3CCl_3 gradient was large

[Prinn *et al.*, 1992, 2000]. We find that the above variations are not determined by any single station.

[11] For comparison to our assumed emissions, we also determined those emissions which would be consistent with a zero trend in OH, by equating the [OH] values at the model grids to their optimally estimated 1978–2004 average values (see auxiliary material). At the times of our inferred OH maxima and minima (Figure 2), the differences between emissions deduced from CH_3CCl_3 observations with constant OH and our reference emissions are outside the range defined by our estimated errors in these emissions (see Figure 3 and auxiliary material).

5. Discussion

[12] The specific variations in OH estimated here are not evident in current theoretical models. To explain the 6% drop in OH in 1997–1999, which is the largest change between successive 3-year estimates inferred here, we examined measured changes in quantities that should affect the production or removal rates of OH. Measurements of carbon monoxide (CO), the major OH sink, indicate anomalously high mole fractions over 1997 to 1999 which have been attributed to massive Indonesian, Russian and North American forest fires at these times [Novelli *et al.*, 2003; Duncan *et al.*, 2003; Dlugokencky *et al.*, 2003; Langenfelds *et al.*, 2002]. The CO and aerosol emissions during the 1997 Indonesian fires alone have been estimated to have lowered global late-1997 OH levels by 6% [Duncan *et al.*, 2003]. Utilizing results from a 3D model [Spivakovsky *et al.*, 2000], we conclude that the above CO increases should have produced a similar OH decrease in 1998. Expected enhancements in (unmeasured) non-methane hydrocarbons due to these fires could amplify these OH decreases, while increases in (unmeasured) NO_x could lower them. Global concentrations of the second major OH sink, CH_4 , were also anomalously high in mid - 1997 to mid - 2000 [Prather and Ehhalt, 2001; Dlugokencky *et al.*, 2003; Langenfelds *et al.*, 2002]. These positive anomalies in CO and CH_4 may, in

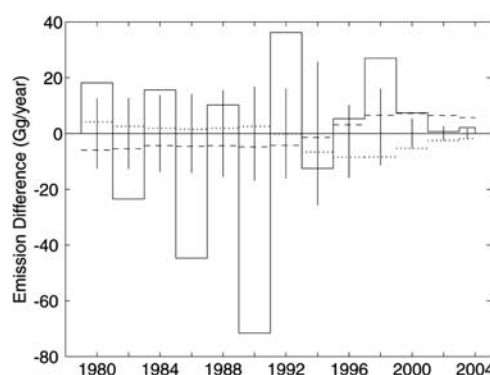


Figure 3. Adjustments (solid lines, 2-year averages) to the “reference” emission estimates (see auxiliary material) needed to yield zero OH trend using AGAGE data. Also shown for comparison are the 2-year averages of the assumed (1σ) error bars on the reference emissions (vertical bars), the Wennberg *et al.* [2004] minus the Yvon-Lewis and Butler [2002] ocean fluxes (dashed lines), and the McCulloch and Midgley [2001] industry-based emissions minus our reference emissions (dotted lines).

combination, help explain our inferred anomalously low global OH concentrations from 1997–1999. Note also that lowering OH by itself leads to positive trends in CO and CH₄ and vice versa [Thompson, 1992]. Large fires also occurred in 1994–1995 and produced about 48% of the CO in the 1997–1998 fires [Novelli et al., 2003; Langenfelds et al., 2002], coinciding approximately with the second largest decrease in successive 3-year OH concentrations reported here (Figure 2). Finally, Yurganov et al. [2004] have reported CO anomalies in 2002 and 2003 that are about 80% of those in 1997–1998 presumably due to large boreal wildfires. However, we see only a slight decrease in our annual-average OH estimates from 2001 to 2003 (Figure 2). This may be due to the CO being emitted more at boreal latitudes where it has less influence on the tropical-weighted OH estimated here, or to the fact that 2002–2003 was not a strong ENSO period.

[13] The ENSO phenomenon could affect both OH production (through water vapor and cloud changes) and OH loss (through temperature changes). As we showed previously [Prinn et al., 2001], at the global level there is tentative evidence for a positive correlation between warm, cloudy El Niño events and low global OH perturbations. Since there was a strong El Niño in 1998, this could have added to the effects of enhanced CO and CH₄ levels to further help explain the 1997–1999 OH low.

[14] The OH concentration in 2003–2004 is indistinguishable (-0.18^{+13}_{-9} %) from that in 1979–1981. Our small derived OH linear trend ($0.2^{+0.8}_{-0.4}$ % year⁻¹) is consistent with interpretation of the observed decelerating 1984–2002 CH₄ trend as an approach to steady-state with a nearly-constant CH₄ lifetime [Dlugokencky et al., 2003]. Our results support the hypothesis that global fires can affect OH chemistry [Duncan et al., 2003], but further studies are needed to test this and identify other possible causes for our reported OH behavior.

[15] **Acknowledgments.** The ALE/GAGE/AGAGE projects involved contributions from many people beyond the authors [see Prinn et al., 2000, 2001]. AGAGE support comes from the National Aeronautics and Space Administration (NASA) with important contributions also from the Department of Environment, Food and Rural Affairs (DEFRA, United Kingdom), Commonwealth Scientific and Industrial Research Organisation (CSIRO, Australia), Bureau of Meteorology (Australia), and the National Oceanic and Atmospheric Administration (NOAA).

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